MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

CHARGE,PROPELLING,5"/54

FLEET ISSUE UNIT LOAD

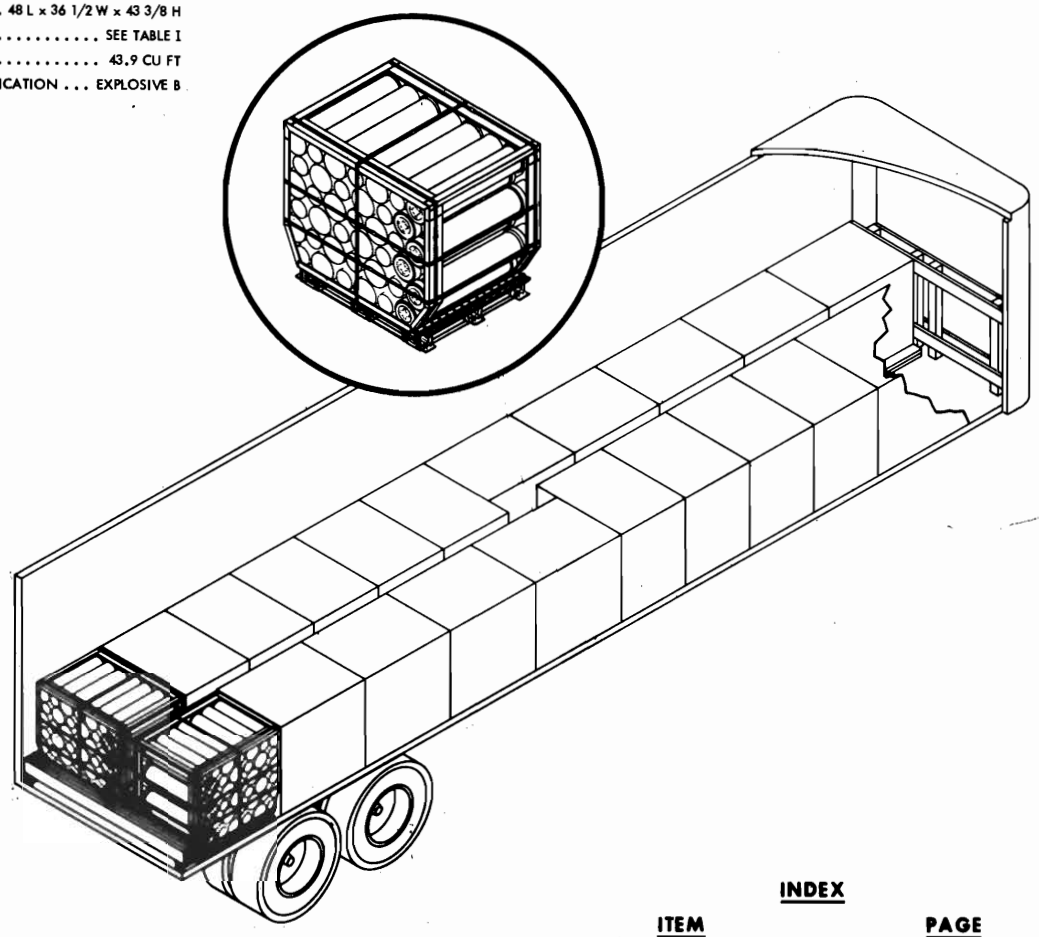
MIL-STD-1320-75 A (NAVY)

25 MARCH 1976

SUPERSEDING
MIL-STD-1320-75 (NAVY)
10 SEPTEMBER 1974

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/5
DIMENSIONS ..... 48 L x 36 1/2 W x 43 3/8 H
WEIGHT ..... SEE TABLE I
CUBE ..... 43.9 CU FT
DOT HAZARD CLASSIFICATION ... EXPLOSIVE B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING PROPELLING CHARGE, 5"/54 IN PALLET ADAPTER MI 16 MOD 0 WHEN PALLETIZED IN ACCORDANCE WITH FLEET ISSUE UNIT LOAD WR-54/5. NO OTHER UNIT LOAD SHALL BE SHIPPED PER THIS DOCUMENT.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS. THE TRAILER SHALL HAVE ITS AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). TRAILER FLOORS MAY BE WOOD, METAL OR METAL WITH WOOD NAILING STRIPS.
3. THE THICKNESS OF THE REAR BLOCKING, DETAILS B AND J, SHALL BE SUFFICIENT TO CAUSE THE BLOCKING TO BEAR AGAINST THE LADING AND THE REAR DOORS WHEN THE DOORS ARE CLOSED.
4. THE LENGTH OF STRUT IN DETAIL F SHALL BE SUFFICIENT TO CAUSE THE REAR BLOCKING ASSEMBLY TO BEAR AGAINST THE LADING AND THE REAR DOORS WHEN THE DOORS ARE CLOSED.
5. THE WIDTH OF SLEEPER, PIECE 7 PAGE 3, SHOULD BE SUFFICIENT TO LIMIT LATERAL MOVEMENT OF UNIT LOADS TO ONE INCH MAXIMUM WHEN POSITIONED BETWEEN PALLET SKIDS. SUBSTITUTE FILLER ASSEMBLY, DETAIL D, FOR SLEEPERS WHEN TRAILER FLOORS ARE NON-NAILABLE.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA. ATTACH "EXPLOSIVE B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L751; NAILS FF-N105, TYPE II, STYLE 10, COMMON BRIGHT.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

TABLE I

TANK	WEIGHT OF UNIT LOAD LBS (APPROX)	NUMBER OF UNIT LOADS FOR FTL		FTL WEIGHT LBS (APPROX)	
		40 & 42 FT TRAILER	44 & 45 FT TRAILER	40 & 42 FT TRAILER	44 & 45 FT TRAILER
MK 14	1683	22	24	37,026	40,392
MK 14 MOD 1	1776	22	24	39,072	42,624
MK 21 MOD 0	1796	22	24	39,512	43,104
REDUCED MK 14	1406	22	24	30,932	33,744

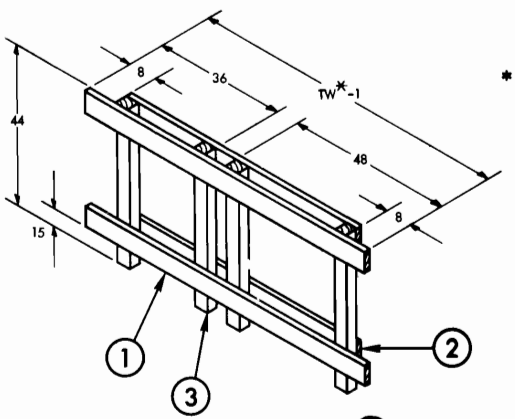
FTL 40 & 42 FT TRAILERS (VAN)

1. A FTL FOR A 40 AND 42 FT TRAILER CONSISTS OF 22 UNIT LOADS. THE UNIT LOADS SHALL BE POSITIONED IN THE TRAILER AND BLOCKED AND BRACED AS SHOWN ON PAGE 3.

WARNING

DO NOT USE TRAILERS WITH ROLL UP DOORS.

2. THE BLOCKING AT THE REAR OF THE LOAD IN A 40 FT TRAILER SHALL BE AS SHOWN ON PAGE 3 AND SHALL COMPLY WITH GENERAL NOTE 3.
3. WHEN BLOCKING THE REAR OF A 42 FT TRAILER, USE THE REAR BLOCKING ASSEMBLY, DETAIL F PAGE 4, AND INSTALL AT REAR OF TRAILER AS SHOWN ON PAGE 5. THE ASSEMBLY SHALL COMPLY WITH GENERAL NOTE 4.
4. WHEN THE TRAILER HAS METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS THAT ARE IMPROPERLY LOCATED TO ACCEPT NAILING, SUBSTITUTE FILLER ASSEMBLY, DETAIL D PAGE 4, FOR SLEEPERS. INSTALL ASSEMBLIES AS SHOWN ON PAGE 5.

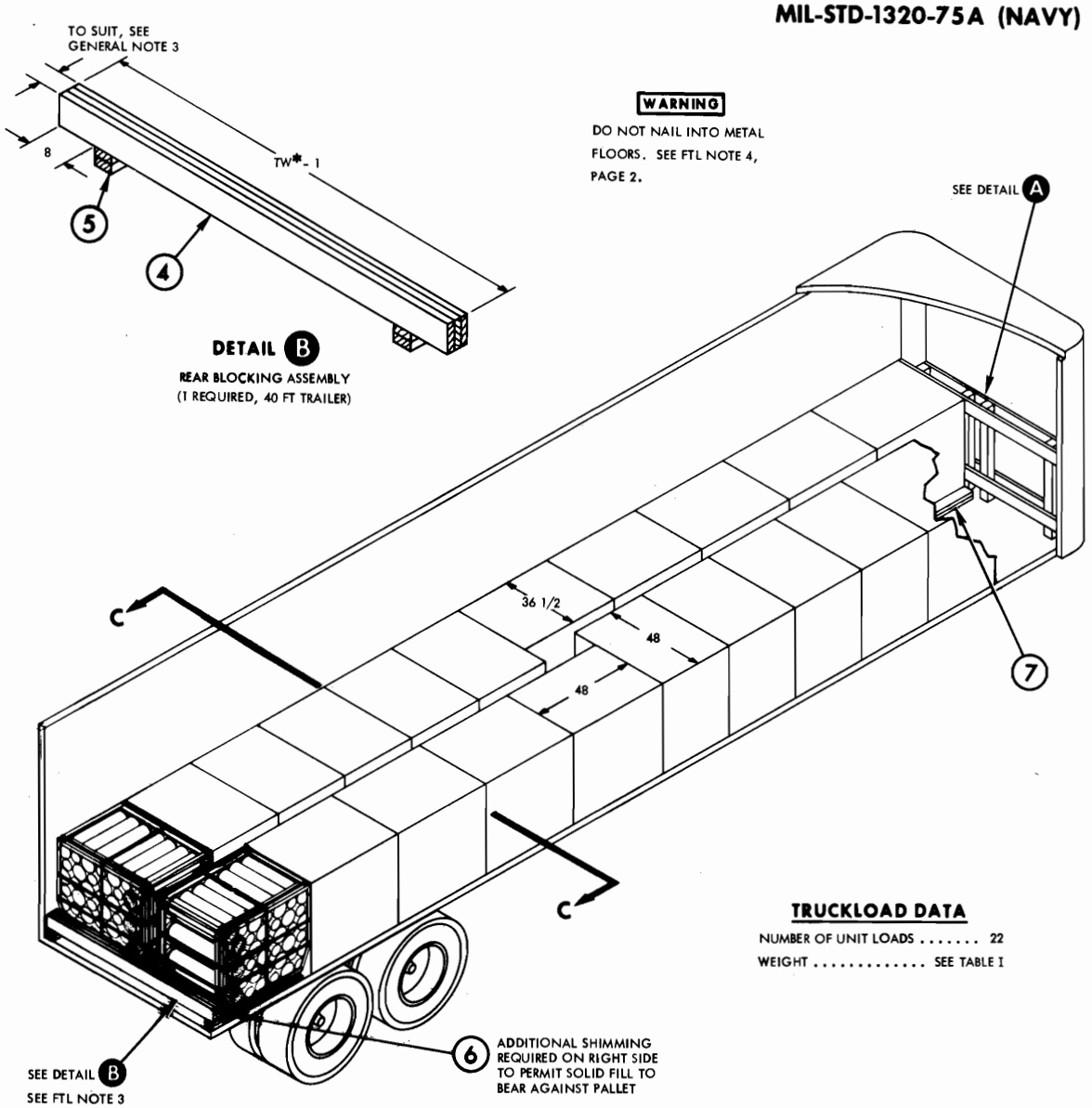


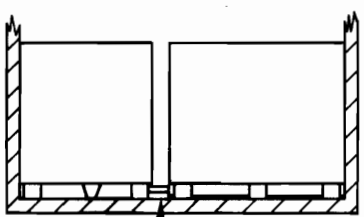
* TW = TRAILER WIDTH

DETAIL A

FRONT BULKHEAD
(1 REQUIRED)

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SECTION C-C

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS AND NAILING DATA						
7	SLEEPER	2 x WIDTH TO SUIT x 48	20	SEE SECTION C-C		
6	SHIM	THICKNESS TO SUIT x 6 x 36 1/2	SEE FIELD NOTE	4	3 PER JOINT	10d
5	SUPPORT	2 x 4 x TO SUIT	4	4	2 PER JOINT	12d
				5	2 PER JOINT	12d
4	SOLID FILL	THICKNESS TO SUIT x 6 x TW*-1	AS REQD	LAMINATE	4 PER JOINT	10d
3	VERTICAL	4 x 4 x 44	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 x 6 x TO SUIT	2	3	2 PER JOINT	12d
1	AFT CROSSMEMBER	2 x 6 x TW*-1	2	3	2 PER JOINT	12d

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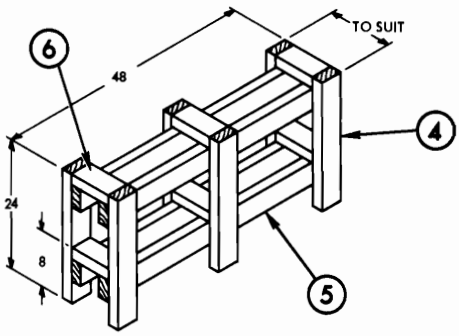
FTL 44 & 45 FT TRAILERS (VAN)

1. A FTL FOR A 44 AND 45 FT TRAILER CONSISTS OF 24 UNIT LOADS. THE UNIT LOADS SHALL BE POSITIONED IN THE TRAILER AND BLOCKED AND BRACED AS SHOWN ON PAGE 5.

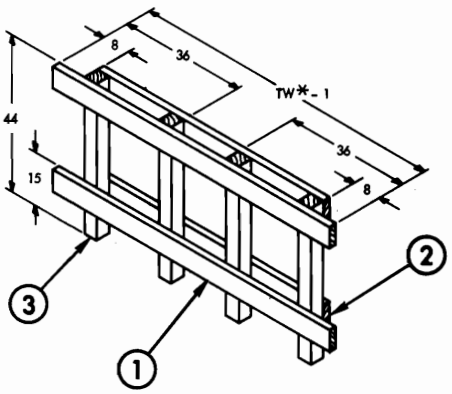
WARNING

TRACTORS USED TO HAUL THIS LOAD
SHALL HAVE A SLIDING FIFTH WHEEL
SO THAT WEIGHT MAY BE SHIFTED
FROM THE DRIVE TO STEERING AXLES.

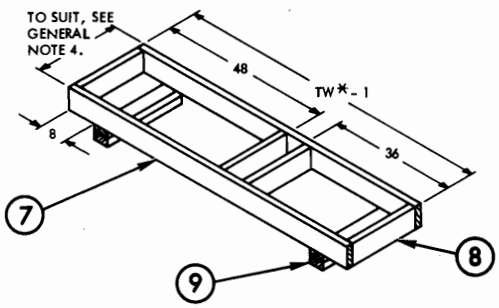
2. THE BLOCKING AT THE REAR OF THE LOAD IN A 45 FT TRAILER SHALL BE AS SHOWN ON PAGE 5. THE ASSEMBLY SHALL COMPLY WITH GENERAL NOTE 4.
3. WHEN BLOCKING THE REAR OF A 44 FT TRAILER, USE THE REAR BLOCKING ASSEMBLY, DETAIL B PAGE 3, AND INSTALL THE ASSEMBLY AS SHOWN ON PAGE 3. THE ASSEMBLY SHALL COMPLY WITH GENERAL NOTE 3.
4. THE LOAD SHOWN ON PAGE 5 SHOWS THE PROPER PROCEDURES FOR TRAILERS WITH NON-NAILABLE FLOORS. IF THE TRAILER HAS A NAILABLE FLOOR, THE PRINCIPLES SHOWN ON PAGE 3 MAY BE APPLIED.



DETAIL D
FILLER ASSEMBLY
(1 REQUIRED)



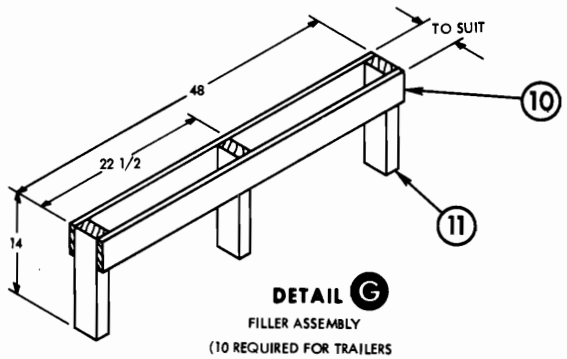
DETAIL E
FRONT BULKHEAD
(1 REQUIRED)



DETAIL F
REAR BLOCKING
ASSEMBLY

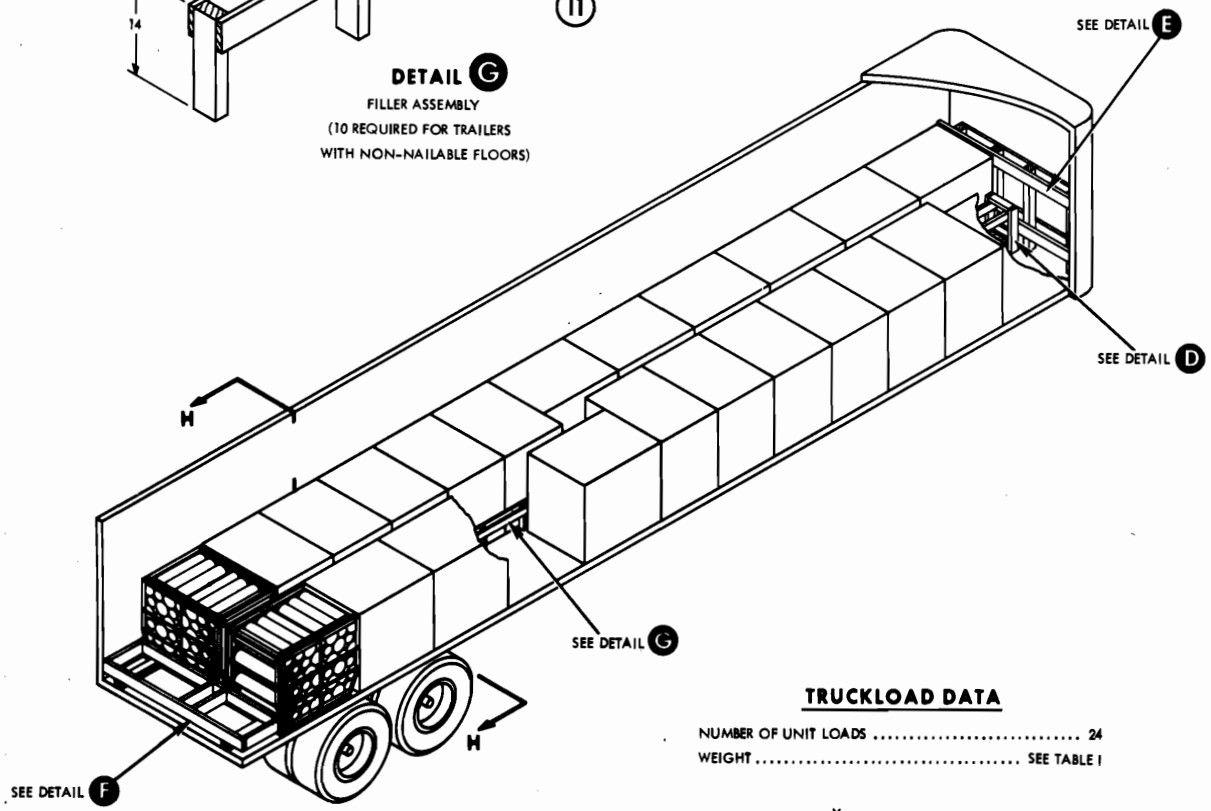
*TW = TRAILER WIDTH

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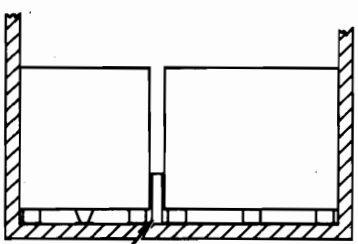
DETAIL G
FILLER ASSEMBLY
(10 REQUIRED FOR TRAILERS
WITH NON-NAILABLE FLOORS)

WARNING
DO NOT USE TRAILERS
WITH ROLL UP DOORS.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 24
WEIGHT ..... SEE TABLE I



SECTION H-H

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
11	VERTICAL	2 X 4 X 14	3	10	-	-
10	LONGITUDINAL	THICKNESS TO SUIT X 4 X 48	AS REQ'D	11	3/JOINT	12d
9	SUPPORT	2 X 4 TO SUIT	4	~~7~~ 9	~~3/JOINT~~ 3	~~10d~~ 10d
8	STRUT	2 X 6 TO SUIT	4	SEE 7	-	-
7	CROSSMEMBER	2 X 6 X TW*-1	2	8	3/JOINT	16d
6	STRUT	2 X 4 TO SUIT	6	5	2/JOINT	12d
5	HORIZONTAL	2 X 4 X 48	4	4	2/JOINT	10d
4	UPRIGHT	2 X 4 X 24	6	SEE 5	-	-
3	VERTICAL	4 X 4 X 44	4	SEE 1 & 2	-	-
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	2/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	2/JOINT	12d

LIST OF MATERIALS & NAILING DATA

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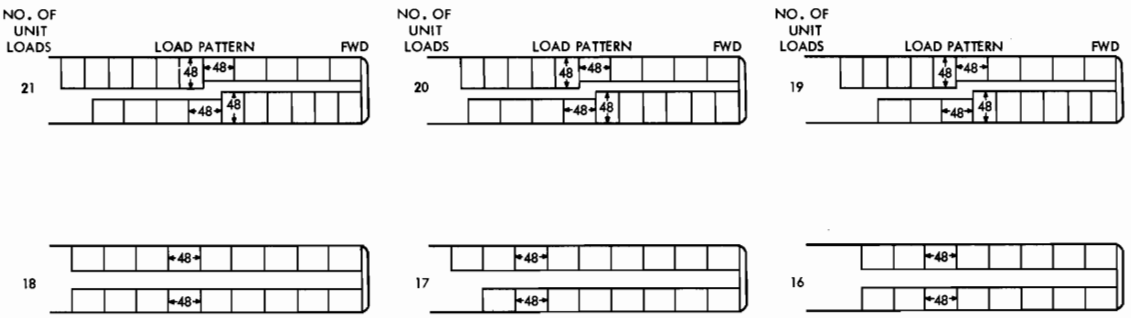
LTL 40 FT TRAILER (VAN)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS SHIPPED FROM TABLE II AND POSITION THE UNIT LOADS ACCORDINGLY.
2. WHEN REQUIRED TO SHIP 21 UNIT LOADS, USE BLOCKING, DETAIL J, AT REAR OF LOAD AND REAR BRACE ASSEMBLY, DETAIL K, FOR SHORT ROW (2 REQUIRED). FOLLOW PRINCIPLES OF LTL ON PAGE 7.
3. WHEN REQUIRED TO SHIP 2 TO 20 UNIT LOADS, USE REAR BRACE ASSEMBLY, DETAIL K, 2 REQUIRED FOR EACH ROW. FOLLOW PRINCIPLES OF LTL ON PAGE 7.
4. THE LTL SHOWN ON PAGE 7 DOES NOT LIMIT THE LTL SHIPMENTS TO THIS QUANTITY. THE PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHOD OF LOADING LTL SHIPMENTS.

WARNING

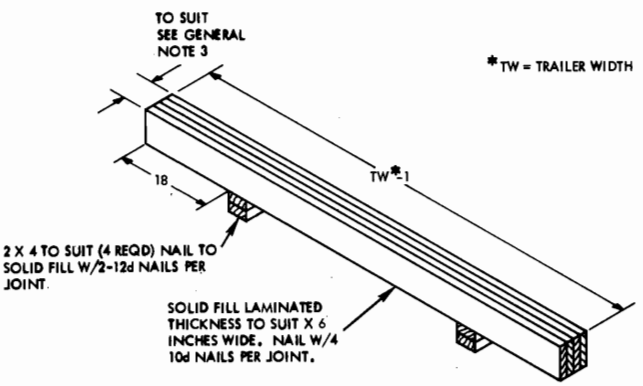
DO NOT USE METAL FLOOR TRAILERS
FOR LTL SHIPMENTS.

TABLE II

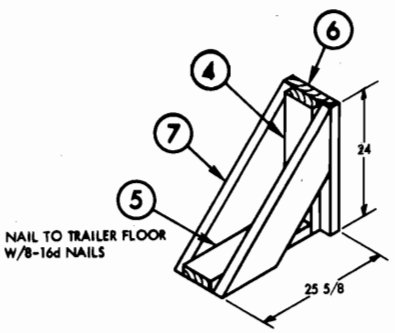


15 UNIT LOADS OR LESS:

POSITION UNIT LOADS IN TRAILER USING A "DOUBLE ROW" PATTERN. TO PREVENT REARWARD MOVEMENT, INSTALL 2 BRACES, DETAIL K, PER ROW AT REAR OF LOAD. (SEE LTL PAGE 7.)

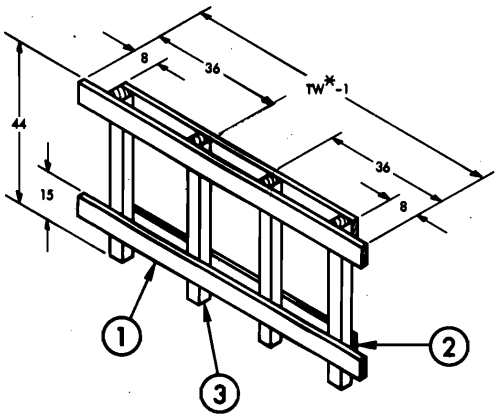


DETAIL J
REAR BLOCKING
(1 REQUIRED)

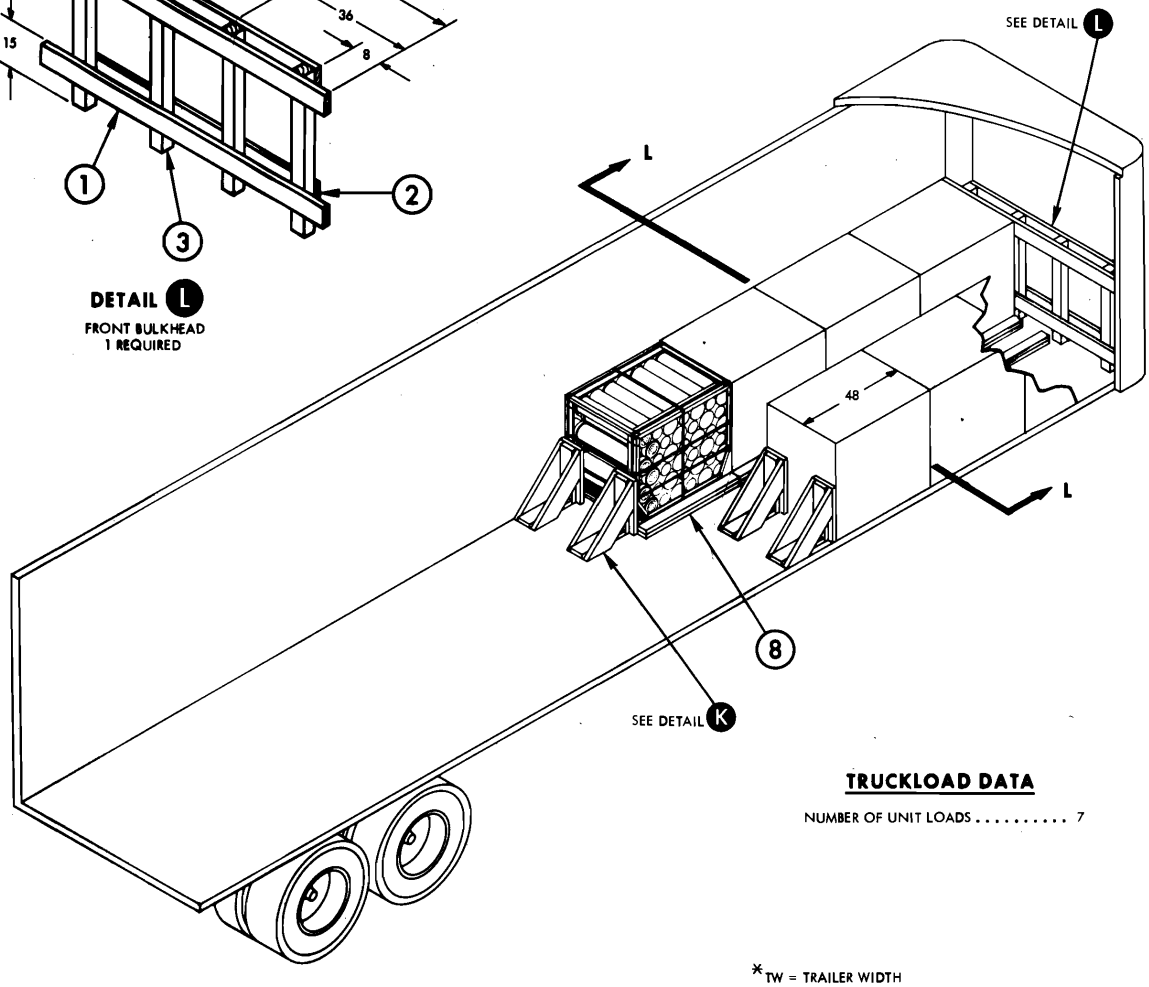


DETAIL K
REAR BRACE ASSEMBLY
(4 REQUIRED)

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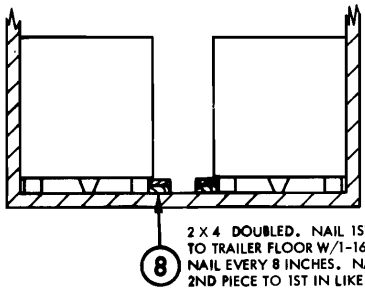
DETAIL L
FRONT BULKHEAD
1 REQUIRED



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 7

*TW = TRAILER WIDTH



PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
8	SLEEPER	2 x 4 x 48	14	SEE SECTION L-L		
7	BRACE DIAGONAL	2 x 6 TO SUIT	8	4 & 5	4 PER JOINT	12d
6	BRACE BACKUP	2 x 10 x 24	4	4	6	10d
5	BRACE CLEAT	2 x 6 x 22 5/8	4	SEE DETAIL K		
4	BRACE UPRIGHT	2 x 6 x 24	4	5	3 PER JOINT	12d
3	VERTICAL	4 x 4 x 44	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 x 6 x TO SUIT	2	3	2 PER JOINT	12d
1	AFT CROSSMEMBER	2 x 6 x TW*-1	2	3	2 PER JOINT	12d

LIST OF MATERIALS AND NAILING DATA

SECTION L-L

LTL 40 FT. TRAILER (VAN)

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N219)

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